

Find the linear classifier for an OR gate using Logistic Regression

The linear equation for a two input OR gate is $\mathbf{w} \cdot \mathbf{x} = w_1x_1 + w_2x_2 + b$. Let us assume $w_1=0.5$, $w_2=0.75$ and $b = -1.25$. So the initial z-score is

$$z = 0.5x_1 + 0.75x_2 - 1.25$$

1 st Cycle			
$x_1=0, x_2=0, y=0$	$\mathbf{w \cdot x + b = -1.25}$	$\tilde{y} = \sigma(\mathbf{w \cdot x + b}) = 0.22$ $hw(\mathbf{x}) = 0$	$y == hw(\mathbf{x})$ w_1, w_2, b remain same
$x_1=0, x_2=1, y=1$	$\mathbf{w \cdot x + b = -0.50}$	$\tilde{y} = \sigma(\mathbf{w \cdot x + b}) = 0.38$ $hw(\mathbf{x}) = 0$	$y \neq hw(\mathbf{x})$ w_1, w_2, b need to updated
		$w_1 = .5 + .2(1-0.38)(0) = 0.5$ $w_2 = .75 + .2(1-0.38)(1) = 0.87$ $b = -1.25 + .2(1-0.38) = -1.13$	
		$z = 0.5x_1 + 0.87x_2 - 1.13$	
$x_1=1, x_2=0, y=1$	$\mathbf{w \cdot x + b = -0.63}$	$\tilde{y} = \sigma(\mathbf{w \cdot x + b}) = 0.35$ $hw(\mathbf{x}) = 0$	$y \neq hw(\mathbf{x})$ w_1, w_2, b need to be updated
		$w_1 = .5 + .2(1-0.35)(1) = 0.63$ $w_2 = .87 + .2(1-0.35)(0) = 0.87$ $b = -1.13 + .2(1-0.35) = -1.0$	
		$z = 0.63x_1 + 0.87x_2 - 1.0$	
$x_1=1, x_2=1, y=1$	$\mathbf{w \cdot x + b = 0.5}$	$\tilde{y} = \sigma(\mathbf{w \cdot x + b}) = 0.62$ $hw(\mathbf{x}) = 1$	$y == hw(\mathbf{x})$ w_1, w_2, b remain same
		$w_1 = 0.63$ $w_2 = 0.87$ $b = -1.0$	
		$z = 0.63x_1 + 0.87x_2 - 1.0$	
2 nd Cycle			

x1=0, x2=0, y=0	w.x+b=-1.03	$\tilde{y} = \sigma(w.x+b) = 0.27$ hw(x)=0	y == hw(x) w1,w2,b remain same
x1=0, x2=1, y=1	w.x+b=-0.13	$\tilde{y} = \sigma(w.x+b) = 0.47$ hw(x)=0	y != hw(x) w1,w2,b need to be updated
		w1 = .63+.2(1-0.47)(0)=0.63 w2=.87+.2(1-0.47)(1)=1.78 b=-1.0+.2(1-0.47) = -0.09	
		z=0.63x1+1.78x2-0.09	
x1=1, x2=0, y=1	w.x+b=0.54	$\tilde{y} = \sigma(w.x+b) = 0.63$ hw(x)=1	y = hw(x) w1,w2,b remain same
		z=0.63x1+1.78x2-0.09	
x1=1, x2=1, y=1	w.x+b=2.32	$\tilde{y} = \sigma(w.x+b) = 0.91$ hw(x)=1	y == hw(x) w1,w2,b remain same
		Z = 0.63x1+1.78x2-0.09	